

Electrostatics

Formal Formula Sheet

Typeset from handwritten notes (pp. 1–5) with standardised notation

Notation and Naming Conventions

Symbol	Meaning (used consistently throughout)
k	Coulomb constant, $k = \frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
q, Q	Charge (q : point/test charge; Q : total charge of a body); $Q = ne$
r	Distance of the field point from the source charge, or from the <i>centre</i> of a sphere/dipole
x	Distance of the field point along the <i>axis</i> of a ring or disk, measured from its centre
R	Radius of a ring, arc, disk, or sphere
d	Separation between two charges (dipole length), or distance from a point charge in potential formulas
ℓ	Length of a line charge
θ	Angle (to the axis for dipoles; to the perpendicular for line charge); α : angle subtended by an arc
λ, σ, ρ	Linear, surface and volume charge density
$\vec{E}, \vec{p}$	Electric field; dipole moment (direction from $-q$ to $+q$)
V, U, W_{EF}	Electric potential; potential energy; work done by the electric force
Φ	Electric flux

1 Basic Relations

Quantity	Formula	Remarks / conditions
Quantization of charge	$Q = ne$	n integer, $e = 1.6 \times 10^{-19} \text{ C}$
Coulomb constant	$k = \frac{1}{4\pi\epsilon_0}$	
Coulomb's force	$F = \frac{k q_1 q_2}{r^2}$	r = separation of the charges
Field of a point charge	$E = \frac{kq}{r^2}$	Directed away from $+q$
Force on a charge in a field	$\vec{F} = q\vec{E}$	
Acceleration of a charge	$a = \frac{qE}{m}$	
Linear charge density	$\lambda = \frac{q}{\ell}$	
Surface charge density	$\sigma = \frac{q}{A}$	

Quantity	Formula	Remarks / conditions
Volume charge density	$\rho = \frac{q}{V}$	

2 Electric Field of Charge Distributions

Quantity	Formula	Remarks / conditions
Ring, on its axis	$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$	Obtained from $\int dE \cos \theta$; x from the centre
Arc subtending angle α , at its centre	$E = \frac{2k\lambda}{R} \sin\left(\frac{\alpha}{2}\right)$	R = radius of arc
Finite line charge (perpendicular component)	$E_{\perp} = \frac{k\lambda}{r} (\sin \theta_1 + \sin \theta_2)$	r = perpendicular distance from the line
Finite line charge (parallel component)	$E_{\parallel} = \frac{k\lambda}{r} (\cos \theta_1 - \cos \theta_2)$	Along the line
Infinite line charge	$E_{\perp} = \frac{2k\lambda}{r}$	Special case $\theta_1 = \theta_2 = 90^\circ$
Semi-infinite line charge	$E = \frac{\sqrt{2}k\lambda}{r}$	Point at perpendicular distance r from the end; field is at 45° to the line
Uniformly charged disk, on its axis	$E = \frac{\sigma}{2\varepsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right]$	R = radius, x = distance from the disk centre
Infinite uniformly charged sheet	$E = \frac{\sigma}{2\varepsilon_0} \hat{n}$	Independent of distance
Infinite non-conducting sheet	$E = \frac{\sigma}{2\varepsilon_0}$	

3 Flux and Gauss's Law

Quantity	Formula	Remarks / conditions
Electric flux	$\Phi = \int \vec{E} \cdot d\vec{A}$	
Gauss's law	$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{in}}}{\varepsilon_0} = \frac{1}{\varepsilon_0} \int \rho dV$	Q_{in} = charge enclosed by the closed surface
Flux through a ring (disk), point charge q on its axis	$\Phi = \frac{q}{2\varepsilon_0}$	Charge placed very close to the ring's plane ($\theta \rightarrow 90^\circ$), so one side receives half the total flux q/ε_0

Quantity	Formula	Remarks / conditions
Flux through a ring (disk), general case	$\Phi = \frac{q}{2\epsilon_0} (1 - \cos \theta)$	θ = half-angle subtended by the ring at q , $\cos \theta = \frac{x}{\sqrt{R^2 + x^2}}$; reduces to $q/2\epsilon_0$ at $\theta = 90^\circ$

4 Field of Spherical Distributions

Quantity	Formula	Remarks / conditions
Solid non-conducting sphere, inside ($r < R$)	$\vec{E} = \frac{\rho \vec{r}}{3\epsilon_0}$	Grows linearly with r
Solid non-conducting sphere, outside ($r > R$)	$E = \frac{kQ}{r^2} \hat{r}$	As if all charge at the centre
Hollow sphere, inside ($r < R$)	$E = 0$	
Hollow sphere, outside ($r > R$)	$E = \frac{kQ}{r^2}$	

5 Electrostatic Pressure and Force

Quantity	Formula	Remarks / conditions
Electrostatic pressure on a charged surface	$P = \frac{\sigma^2}{2\epsilon_0}$	σ = surface charge density; acts outward on the surface
Force from electrostatic pressure (general)	$\vec{F} = \int P \, d\vec{A}$	$d\vec{A}$ = area vector, directed along the outward normal
Force on a flat area, uniform P	$F = PA \cos \theta$	θ = angle between $\vec{A}$ and the direction of the force component required

6 Potential Energy and Work

Quantity	Formula	Remarks / conditions
Change in potential energy	$\Delta U = -W_{\text{conservative}}$	

Quantity	Formula	Remarks / conditions
Work by electric force	$W_{EF} = \int \vec{F}_E \cdot d\vec{r} = -\Delta U = U_i - U_f$	
Potential energy of two point charges	$U = \frac{k q_1 q_2}{r}$	r = separation
Energy of a dipole in a field	$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$	

7 Electric Potential

Quantity	Formula	Remarks / conditions
Potential difference	$\Delta V = \frac{\Delta U}{q} = -\frac{W_{EF}}{q}$	
Energy from potential	$\Delta U = q \Delta V$	
Point charge	$V = \frac{kq}{r}$	
Multiple charges	$V = k \sum_i \frac{q_i}{r_i}$	Equals kQ_{total}/r when all charges are at the same distance
Potential from field	$\Delta V = - \int \vec{E} \cdot d\vec{r}$	From $q \Delta V = -W_{EF}$
Field from potential	$E = -\frac{dV}{dr}$	
Uniformly charged ring, on its axis	$V = \frac{kQ}{\sqrt{R^2 + x^2}}$	$\sqrt{R^2 + x^2}$ = distance of the point from the ring
Hollow shell, $r \leq R$	$V = \frac{kQ}{R}$	Constant inside
Hollow shell, $r \geq R$	$V = \frac{kQ}{r}$	
Solid non-conducting sphere, $r \leq R$	$V = \frac{kQ(3R^2 - r^2)}{2R^3}$	
Solid non-conducting sphere, $r \geq R$	$V = \frac{kQ}{r}$	
Equipotential surface	$\Delta V = 0, \quad W_{EF} = 0$	No work is done moving a charge along it

8 Electric Dipole

Quantity	Formula	Remarks / conditions
Dipole moment	$p = q d$	d = distance between the two charges; $Q_{\text{total}} = 0$
Axial field	$E_{\text{axial}} = \frac{2kp}{r^3}$	r = distance from the centre of the dipole
Equatorial field	$E_{\text{eq}} = \frac{kp}{r^3}$	
Field at general point	$E = \frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$	θ measured from the axis; reduces to E_{axial} at $\theta = 0$
Net force in uniform field	$F_{\text{net}} = 0$	
Torque in uniform field	$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta$	
Potential of a dipole	$V = \frac{kp \cos \theta}{r^2} = \frac{k \vec{p} \cdot \vec{r}}{r^3}$	$V = 0$ on the equatorial line